

Vadivelan Murugesan

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EDUCATION

New York University

M.S. Mechatronics and Robotics

New York, USA

2025

SKILLS

Technical Skills: Programming (C, Python, Lua), Computer Vision (OpenCV), Simulation Tools (MATLAB, Simulink, Coppeliassim, Gazebo), Robotics (ROS, Control Systems, Kinematics, Soldering), Embedded Firmware

WORK EXPERIENCE

Surinova Pvt Ltd

Robotics Design Engineer

Chennai, India

11/2022 – 07/2023

- Collaborated on the development of a tea-harvesting rover by integrating ODrive motor controllers with STM32F4 microcontrollers and custom PCB hardware, utilizing CAN, SPI, and I2C protocols for real-time communication and control.
- Proficient in Python and C/C++, applied these languages in developing and debugging embedded firmware.
- Utilized advanced testing tools like Picoscope for debugging microcontroller-based projects, resulting in a 50% decrease in troubleshooting time and increased project turnaround efficiency.

Cologenes Healthcare Pvt Ltd

Technical Intern

Salem, India

06/2021 - 08/2021

- Devised an automated segment of collagen sheet production utilizing a laser cutter, reducing processing time by 30% and increasing overall efficiency through optimized computing processes.
- Implemented computer vision techniques to ensure precise automation of collagen sheet production, improving alignment accuracy by 80% and enhancing production throughput by 20%.

e-Yantra Summer Internship, IIT Bombay

Research Project Intern

Bombay, India

05/2021 - 06/2021

- Collaborated with the project lead and team to leverage simulation environments for designing effective solutions to real-world challenges.
- Conducted in-depth exploration of Blue Zero middleware and CoppeliaSim plugins, assessing their potential for seamless hardware implementation.

PROJECTS

Robot Perception: Solving a 3D Maze Using Graph Networks

2024

- Developed a Spatial Vicinity Graph, treating each point in the 3D maze as a node and connecting them based on global image descriptions. The graph ensured the shortest and most efficient pathfinding with 100% accuracy.
- Utilized SuperPoint+SuperGlue for feature extraction and matching, coupled with geometric verification, to identify the destination node within the maze for precise navigation.
- Explored multiple methods, including COLMAP, Pose Graph Optimization with GTSAM, and Stella VSLAM, before refining the final approach, combining vision-based algorithms and graph theory for optimal results.

Key Projects in Perception, Machine Learning, and 3D Mapping ([Github Link](#))

2024

- Developed custom algorithms for RANSAC Plane Fitting and Iterative Closest Point (ICP) to process and align 3D point clouds, achieving precise plane fitting and point cloud registration.
- Built and trained an autoencoder on the Fashion-MNIST dataset for dimensionality reduction and implemented a supervised classifier, achieving 85% accuracy with comprehensive evaluation.
- Created Augmented Reality visualizations using OpenCV and Aruco markers, projecting 3D objects onto real-world tags with precise calibration and rendering techniques.
- Implemented object tracking in dynamic video sequences using YOLOv8 and tracking algorithms, ensuring robust and persistent identification across frames.

Automated Object Retrieval Mobile Robot ([Github Link](#))

2024

- Developed an automated mobile robot system with integrated camera and Raspberry Pi for real-time object identification using OpenCV. Designed perception algorithms for object detection and localization.
- Engineered and programmed a mobile robot with a 2-DOF arm, achieving manipulation capabilities to autonomously navigate and relocate objects within an arena, demonstrating advanced control and planning capabilities in dynamic environments.

Color Cube Sorting Robotic Arm ([Github Link](#))

2024

- Engineered a 4-DOF robotic arm with a gripper to sort color cubes on a conveyor belt, integrating a color sensor for real-time identification and an ultrasonic sensor for detecting cube proximity.
- Enhanced the automation process by utilising cogs in Parallax Propeller Board, implementing multiprocessing that enables simultaneous colour and distance sensing, improving the efficiency and accuracy of the sorting mechanism.

Teleoperation of a 4-DOF Robotic Arm ([Github Link](#))

2024

- Designed and prototyped a 4DOF Robotic Arm, programmable through a Puppet model with 4 potentiometers, closely resembling industrial pick and place machines.
- Implemented a system allowing the robot to save and cycle through up to 10 reprogrammable positions, enhancing operational flexibility.
- Developed a real-time mirror mode that replicates the puppet robot's orientation, demonstrating advanced real-time control capabilities.